

Quiz (Carolina Reaper)

Q1. What is the expansion of $(a + b)^2$?

- (A) $a^2 + 2ab + b^2$
(B) $a^2 - 2ab + b^2$
(C) $a^2 + ab + b^2$
(D) $a^2 - ab + b^2$
(E) $a^2 + b^2$

Q2. Which of the following is a perfect square trinomial?

- (A) $x^2 + 4x + 4$
(B) $x^2 - 4x - 4$
(C) $x^2 + 4x - 4$
(D) $x^2 - 4x + 4$
(E) $x^2 + 4$

Q3. What is the factored form of $x^2 + 6x + 9$?

- (A) $(x + 3)^2$
(B) $(x - 3)^2$
(C) $(x + 3)(x - 3)$
(D) $(x + 2)(x + 4)$
(E) $(x - 2)(x - 4)$

Q4. What is the expansion of $(a - b)^2$?

- (A) $a^2 + 2ab + b^2$
(B) $a^2 - 2ab + b^2$
(C) $a^2 + ab + b^2$
(D) $a^2 - ab + b^2$
(E) $a^2 + b^2$

Q5. Which of the following is not a perfect square trinomial?

- (A) $x^2 + 4x + 4$
(B) $x^2 - 4x + 4$
(C) $x^2 + 4x - 4$
(D) $x^2 - 4x - 4$
(E) $x^2 + 4$

Q6. What is the factored form of $x^2 - 8x + 16$?

- (A) $(x + 4)^2$
(B) $(x - 4)^2$
(C) $(x + 3)(x - 3)$
(D) $(x + 2)(x + 4)$
(E) $(x - 2)(x - 4)$

Q7. What is the expansion of $(2x + 3)^2$?

- (A) $4x^2 + 12x + 9$
(B) $4x^2 - 12x + 9$
(C) $4x^2 + 6x + 9$
(D) $4x^2 - 6x + 9$
(E) $4x^2 + 9$

Q8. Which of the following is the factored form of $x^2 + 10x + 25$?

- (A) $(x + 5)^2$
(B) $(x - 5)^2$
(C) $(x + 3)(x + 7)$
(D) $(x + 2)(x + 8)$
(E) $(x - 2)(x - 8)$

Open Ended Questions (FRQ)**Q9.** Factor the expression $x^2 + 12x + 36$.**Q10.** Expand the expression $(3x - 2)^2$ and simplify.

Chef's Tips

- Emphasize the importance of recognizing special patterns when expanding binomials.
- Make sure students understand the relationship between perfect square trinomials and their factored forms.
- Encourage students to check their work by multiplying the factored form to ensure it equals the original expression.